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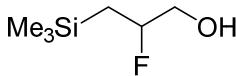
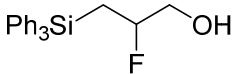
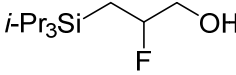
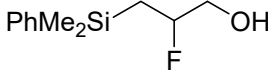
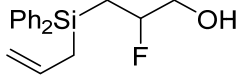
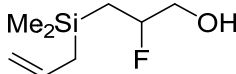
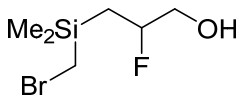
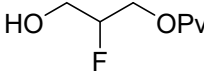
Regioselective Fluorohydrin Synthesis from Allylsilanes

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General: All reactions were carried out in Teflon vials in air unless otherwise noted. IR: Nicolet iS10 spectrometer, wavenumbers ($\tilde{\nu}$) in cm⁻¹. All reagents were purchased and used as received unless otherwise mentioned. TLC analysis used 0.25 mm silica layer fluorescence UV₂₅₄ plates. Flash chromatography: SilaCycle silica gel P60 (230-400 mesh). NMR: Spectra were recorded on a Varian Mercury 300, or Inova 500 spectrometer in the solvents indicated; chemical shifts (δ) are given in ppm, coupling constants (J) in Hz. The solvent signals

were used as references (residual CHCl_3 in CDCl_3 : $\delta_{\text{H}} \equiv 7.26$ ppm, $\delta_{\text{C}} \equiv 77.0$ ppm; residual C_6H_6 in C_6D_6 : $\delta_{\text{H}} \equiv 7.15$ ppm, $\delta_{\text{C}} \equiv 128.0$ ppm).

General Procedure (A1): Oxone epoxidation of allylsilanes. To a mixture of the allylsilane (1.0 mmol) and *tert*-butyl ammonium hydrogen sulfate (0.0136 g, 4 mol%) in acetonitrile-dimethoxymethane (2:1) (8 mL), acetone (2.2 mL, 30 mmol) was added aqueous K_2CO_3 (0.1M) (0.2 mmol). With vigorous stirring, a solution of oxone (3 mmol, in 8 mL of 4×10^{-4} M EDTA solution) and aqueous K_2CO_3 (13.3 mmol, in 8 mL of H_2O) were separately added via syringe pump over the indicated time. Upon completion, the reaction mixture was extracted with hexanes (3 x 20 mL). The combined extracts were washed with brine, dried with MgSO_4 , and filtered before removing the solvent under reduced pressure. The crude product was used directly in the next reaction without further purification.

General Procedure (A2): $\text{HF} \cdot \text{Et}_3\text{N}$ epoxide ring opening. A Teflon vial was charged with the epoxysilane (1 equiv) and DCM (0.05-0.1 M). The mixture was stirred and $\text{HF} \cdot \text{Et}_3\text{N}$ (5 equiv) was added dropwise via syringe. The reaction vessel was sealed with a Teflon screw cap and the solution was stirred over the indicated time at room temperature. The mixture was then poured into a beaker containing saturated aqueous NaHCO_3 (75 mL) and allowed to stir until no evolution of CO_2 was observed. The mixture was transferred to a separatory funnel and extracted with CH_2Cl_2 (3 x 25 mL). The combined organic extracts were dried with MgSO_4 and filtered before removing the solvent under reduced pressure. The crude product was then purified by flash column chromatography on silica.

General Procedure (B): One-pot fluorohydrin synthesis from allylsilanes. This procedure was adapted from that previously described by Sedgwick et al.:¹ A Teflon vial was charged with mCPBA (1.3 equiv) and DCM (0.0625-0.1 M) and stirred until the mCPBA had dissolved. With stirring, $\text{HF} \cdot \text{Et}_3\text{N}$ (5-7 equiv) was then added followed immediately by the allylsilane (1 equiv) via syringe. The reaction vessel was sealed with a Teflon screw cap and the solution was stirred for the indicated time at room temperature. The mixture was then poured into a beaker containing saturated aqueous NaHCO_3 (75 mL) and allowed to stir until no evolution of CO_2 was observed. The mixture was transferred to a separatory funnel and extracted with CH_2Cl_2 (3 x 25 mL). The combined organic extracts were dried with MgSO_4 and filtered before removing the solvent under reduced pressure. The crude product was then purified by flash column chromatography on silica.

General Shi Epoxidation Procedure (Compounds 1-3). To a mixture of the allylsilane (1.0 mmol) and *tert*-butyl ammonium hydrogen sulfate (0.0136 g, 4 mol%) in acetonitrile (15 mL), shi catalyst (77.5 mg, 0.3 mmol) was added buffer (0.05 M $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10 \text{H}_2\text{O}$ in 4×10^{-4} M aqueous $\text{Na}_2(\text{EDTA})$ (10 mL, 0.1 M). With vigorous stirring, a solution of oxone (1.38 mmol, in 6.5 mL of 4×10^{-4} M $\text{Na}_2(\text{EDTA})$ solution) and aqueous K_2CO_3 (5.8 mmol, in 6.5 mL of H_2O) were separately added via syringe pump over the indicated time. Upon completion, the reaction mixture was extracted with hexanes (3 x 20 mL). The combined extracts were washed with brine, dried with MgSO_4 , and filtered before removing the solvent under reduced pressure. The crude product was used directly in the next reaction without further purification using procedure A2.

(S)-3-(Triisopropylsilyl)propene oxide. $[\alpha]_{\text{D}}^{25} = -5.66$ (*c* 1.45, CH_2Cl_2). NMR spectral data identical

¹ Sedgwick, D. M.; López, I.; Román, R.; Kobayashi, N.; Okoromoba, O. E.; Xu, B.; Hammond, G. B.; Barrio, P.; Fustero, S. *Org. Lett.* **2018**, *20*, 2338–2341.

2-fluoro-3-(trimethylsilyl)propan-1-ol

From allyltrimethylsilane (114.3 mg, 1 mmol) following procedure B was added HF•Et₃N (0.27 mL, 5 mmol, 5 equiv), DCM (16 mL, 0.0625 M) and stirred for 1 hour. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.34) to yield the title compound as a colorless liquid (98.5 mg, 0.65 mmol, 65%).

¹H NMR (500 MHz, CDCl₃) δ 4.79 (ddtd, *J* = 50.1, 8.9, 6.4, 2.7 Hz, 1H), 3.74 – 3.59 (m, 2H), 1.09 (ddd, *J* = 14.5, 13.6, 8.7 Hz, 1H), 0.89 (ddd, *J* = 35.2, 14.5, 6.3 Hz, 1H), 0.09 (d, *J* = 0.8 Hz, 8H).

¹³C NMR (126 MHz, CDCl₃) δ 93.74 (d, *J* = 166.2 Hz), 67.30 (d, *J* = 22.8 Hz), 19.82 (d, *J* = 22.6 Hz), -1.00.

¹⁹F NMR (470 MHz, CDCl₃) δ -176.05 (ddddd, *J* = 49.3, 34.6, 28.7, 20.4, 13.7 Hz).

2-fluoro-3-(triphenylsilyl)propan-1-ol

From allyltriphenylsilane (300.5 mg, 1 mmol) following procedure A1, the reaction was allowed to proceed for 16 hours. This was then taken directly into procedure A2 without purification: Epoxysilane (63.3 mg, 0.2 mmol), HF•Et₃N (0.06 mL, 1 mmol, 5 equiv), DCM (2.5 mL, 0.08 M), stirred for 72 hours. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.2) to yield the title compound as a white solid (53.5 mg, 0.16 mmol, 65%).

¹H NMR (500 MHz, CDCl₃) δ 7.56 – 7.53 (m, 5H), 7.44 – 7.36 (m, 10H), 4.92-4.76 (ddtd, *J* = 49.2, 8.4, 6.4, 2.9 Hz, 1H), 3.67 – 3.51 (m, 2H), 2.01 (ddd, *J* = 15.1, 13.6, 8.2 Hz, 1H), 1.74 (ddd, *J* = 32.6, 14.8, 6.4 Hz, 1H).

¹³C NMR (126 MHz, CDCl₃) δ 135.63, 134.01, 129.80, 128.06, 92.86 (d, *J* = 168.3 Hz), 66.89 (d, *J* = 22.6 Hz), 16.98 (d, *J* = 22.3 Hz).

¹⁹F NMR (470 MHz, CDCl₃) δ -173.21 – -173.55 (m).

HRMS (ESI+) Calculated for C₂₁H₂₁FN₁OSi (M + Na): 359.123791. Found 359.124384

2-fluoro-3-(triisopropylsilyl)propan-1-ol

From allyltriisopropylsilane (198.4 mg, 1 mmol) following procedure A1 the reaction was allowed to proceed for 4 hours. This was then taken directly into procedure A2 without purification: Epoxysilane (60 mg, 0.28 mmol), HF•Et₃N (0.08 mL, 1.4 mmol, 5 equiv), DCM (5.6 mL, 0.05 M) stirred for 1.5 hours. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.4) to yield the title compound as a colorless liquid (63.7 mg, 0.27 mmol, 92%).

¹H NMR (500 MHz, CDCl₃) δ 4.82 (ddddd, *J* = 49.7, 9.8, 7.2, 4.9, 2.7 Hz, 1H), 3.76 – 3.58 (m, 2H), 1.15 – 1.01 (m, 23H), 0.86 (ddd, *J* = 41.6, 14.9, 4.8 Hz, 1H).

¹³C NMR (126 MHz, CDCl₃) δ 93.33 (d, *J* = 166.2 Hz), 67.84 (d, *J* = 23.5 Hz), 18.72, 18.69, 12.19 (d, *J* = 25.1 Hz), 11.25.

¹⁹F NMR (470 MHz, CDCl₃) δ -175.25 – -175.63 (m).

3-(dimethyl(phenyl)silyl)-2-fluoropropan-1-ol

From allyldimethyl(phenyl)silane (176.3 mg, 1 mmol) following procedure A1 the reaction was allowed to proceed for 4 hours. This was then taken directly into procedure A2 without purification: Epoxysilane (192.3 mg, 1 mmol), HF•Et₃N (0.27 mL, 5 mmol 5 equiv), DCM (20 mL, 0.05 M) stirred for 4 hours. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.2) to yield the title compound as a colorless liquid (153.8 mg, 0.724 mmol, 72%).

¹H NMR (500 MHz, CDCl₃) δ 7.56 – 7.48 (m, 2H), 7.42 – 7.32 (m, 3H), 4.81-4.65 (ddtd, *J* = 50.0, 9.0, 6.4, 2.8 Hz, 1H), 3.67 – 3.51 (m, 2H), 1.81 (s, 1H), 1.32 (ddd, *J* = 14.6, 13.3, 8.7 Hz, 1H), 1.12 (ddd, *J* = 35.1, 14.6, 6.1 Hz, 1H), 0.38 (s, 3H), 0.36 (s, 3H).

¹³C NMR (126 MHz, CDCl₃) δ 138.02, 133.50, 129.29, 127.97, 93.43 (d, *J* = 166.6 Hz), 67.14 (d, *J* = 22.8 Hz), 19.25 (d, *J* = 22.7 Hz), -2.14, -2.64.

¹⁹F NMR (470 MHz, CDCl₃) δ -175.73 (ddddd, *J* = 49.0, 34.7, 28.0, 21.1, 13.3 Hz).

3-(allyldiphenylsilyl)-2-fluoropropan-1-ol

From diallyldiphenylsilane (238 mg, 0.9 mmol) following procedure B was added HF•Et₃N (0.27 mL, 4.5 mmol, 5 equiv), DCM (9 mL, 0.1 M) and stirred for 10 hours. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.17) to yield the title compound as a colorless liquid (143 mg, 0.476 mmol, 53%).

¹H NMR (500 MHz, CDCl₃) δ 7.54 (ddt, *J* = 12.4, 7.9, 1.5 Hz, 4H), 7.39 (ddddd, *J* = 13.5, 8.0, 6.8, 4.7 Hz, 6H), 5.79 (ddddd, *J* = 18.2, 15.8, 7.9, 1.2 Hz, 1H), 4.95 (dt, *J* = 16.9, 1.6 Hz, 1H), 4.91 (ddd, *J* = 10.0, 2.2, 1.0 Hz, 1H), 4.85-4.69 (ddtd, *J* = 49.2, 9.0, 6.0, 3.0 Hz, 1H), 3.66 – 3.52 (m, 2H), 2.20 (d, *J* = 7.9 Hz, 2H), 1.72 (ddd, *J* = 14.9, 12.5, 8.8 Hz, 1H), 1.44 (ddd, *J* = 35.5, 14.8, 5.9 Hz, 1H).

¹³C NMR (126 MHz, CDCl₃) δ 134.94, 134.90, 134.43, 134.24, 133.46, 129.77, 129.72, 128.05, 127.98, 115.09, 92.93 (d, *J* = 167.2 Hz), 67.00 (d, *J* = 22.9 Hz), 21.05, 15.76 (d, *J* = 23.2 Hz).

¹⁹F NMR (470 MHz, CDCl₃) δ -175.23 (ddddd, *J* = 48.7, 34.6, 27.5, 21.6, 12.8 Hz).

HRMS (ESI+) Calculated for C₁₈H₂₁FN₃OSi (M + Na): 323.123791. Found 323.124277

3-(allyldimethylsilyl)-2-fluoropropan-1-ol

From diallyldimethylsilane (70.2 mg, 0.5 mmol) following procedure B was added HF•Et₃N (0.135 mL, 2.5 mmol, 5 equiv), DCM (8 mL, 0.0625 M) and stirred for 1 hour. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, R_f ~ 0.34) to yield the title compound as a colorless liquid (50 mg, 0.28 mmol, 56%).

IR (ATR) 2955, 1732, 1606, 1436, 1265, 1170, 830, 734 cm⁻¹.

¹H NMR (500 MHz, CDCl₃) δ 5.77 (ddt, *J* = 16.6, 10.3, 8.1 Hz, 1H), 4.91 – 4.68 (m, 3H), 3.74 – 3.56 (m, 2H), 1.58 (dt, *J* = 8.1, 1.3 Hz, 2H), 1.09 (ddd, *J* = 14.6, 12.7, 9.1 Hz, 1H), 0.88 (ddd, *J* = 37.2, 14.6, 5.9 Hz, 1H), 0.08 (d, *J* = 0.9 Hz, 6H).

¹³C NMR (126 MHz, CDCl₃) δ 134.40, 113.48, 93.49 (d, *J* = 165.9 Hz), 67.33 (d, *J* = 23.1 Hz), 23.56, 17.93 (d, *J* = 23.6 Hz), -3.01, -3.14.

¹⁹F NMR (470 MHz, CDCl₃) δ -176.55 (ddddd, *J* = 49.5, 37.3, 28.5, 20.2, 12.6 Hz).

HRMS (ESI+) Calculated for C₈H₁₇FN₂OSi (M + Na): 199.092491. Found 199.092151

3-((bromomethyl)dimethylsilyl)-2-fluoropropan-1-ol

From allyl(bromomethyl)dimethylsilane (193.2 mg, 1 mmol) following procedure A1 the reaction was allowed to proceed for 4 hours. This was then taken directly into procedure A2 without purification: Epoxysilane (104.6 mg, 0.5 mmol), HF•Et₃N (0.13 mL, 2.5 mmol, 5 equiv), DCM (10 mL, 0.05 M) stirred for 2 hours. The crude product was purified by flash column chromatography on silica (10:1 hexanes:ethyl acetate, *R_f* ~ 0.7) to yield the title compound as a colorless liquid (68.8 mg, 0.3 mmol, 60%).

¹H NMR (500 MHz, CDCl₃) δ 4.89-4.72 (ddddd, *J* = 49.5, 9.6, 6.7, 5.0, 2.7 Hz, 1H), 3.75 – 3.58 (m, 2H), 2.52 (d, *J* = 2.5 Hz, 2H), 1.22 (ddd, *J* = 14.8, 11.1, 9.8 Hz, 1H), 1.02 (ddd, *J* = 40.6, 14.9, 5.0 Hz, 1H), 0.22 (s, 6H).

¹³C NMR (126 MHz, CDCl₃) δ 93.24 (d, *J* = 166.2 Hz), 67.36 (d, *J* = 24.4 Hz), 17.48 (d, *J* = 25.1 Hz), 17.08, -3.21, -3.23.

¹⁹F NMR (470 MHz, CDCl₃) δ -177.82 (dtdd, *J* = 49.8, 40.3, 20.8, 11.2 Hz).

HRMS (ESI+) Calculated for C₆H₁₄BrFNaOS (M + Na): 250.987353. Found 250.988441

2-fluoro-3-hydroxypropyl pivalate (6)

To a solution of 3-(dimethyl(phenyl)silyl)-2-fluoropropan-1-ol (212.34 mg, 0.956 mmol) in DCM (4.8 mL, 0.2 M) at 0 °C was added pyridine (0.23 mL, 2.9 mmol) and pivaloyl chloride (0.28 mL, 2.3 mmol). The mixture was allowed to slowly warm to room temperature while stirring over 16 hr. The reaction was quenched with aq. NaHCO₃ (30 mL) and extracted with DCM (3 x 20 mL). The combined organic extracts dried with MgSO₄ and filtered before removing the solvent under reduced pressure. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, *R_f* ~ 0.7) to yield the compound the pivaloyl ester as a colorless liquid (242.7 mg, 0.819 mmol, 85%).

¹H NMR (500 MHz, CDCl₃) δ 7.54 – 7.50 (m, 2H), 7.39 – 7.36 (m, 3H), 4.80 (ddddd, *J* = 49.1, 9.3, 6.9, 5.7, 2.5 Hz, 1H), 4.14 (ddd, *J* = 27.6, 12.4, 2.5 Hz, 1H), 4.03 (ddd, *J* = 20.7, 12.4, 6.8 Hz, 1H), 1.32 (ddd, *J* = 14.8, 12.4, 9.2 Hz, 1H), 1.21 (s, 9H), 1.14 (ddd, *J* = 36.5, 14.6, 5.7 Hz, 1H), 0.38 (d, *J* = 0.9 Hz, 3H), 0.36 (d, *J* = 0.9 Hz, 3H).

^{13}C NMR (126 MHz, CDCl_3) δ 178.24, 137.91, 133.51, 129.30, 127.97, 90.08 (d, $J = 170.9$ Hz), 67.44 (d, $J = 22.9$ Hz), 38.83, 27.15, 19.63 (d, $J = 23.6$ Hz), -2.18, -2.65.

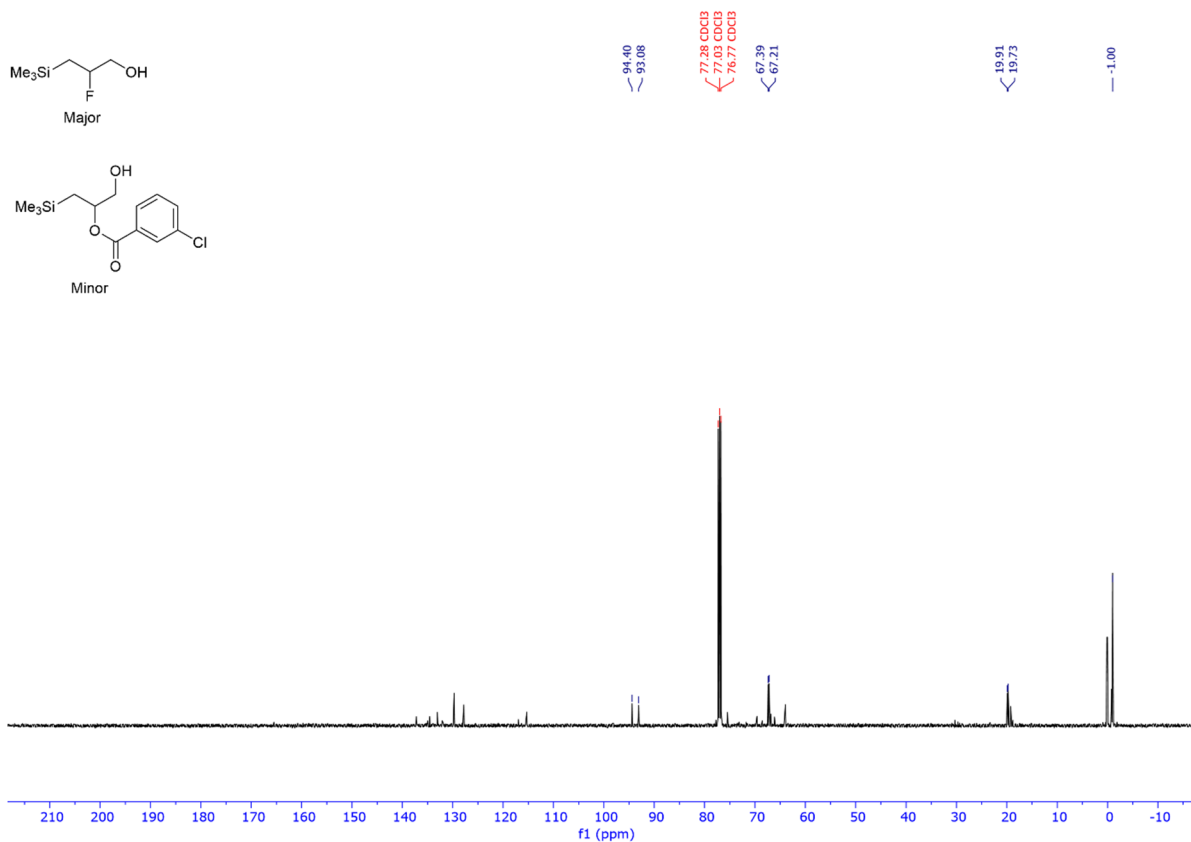
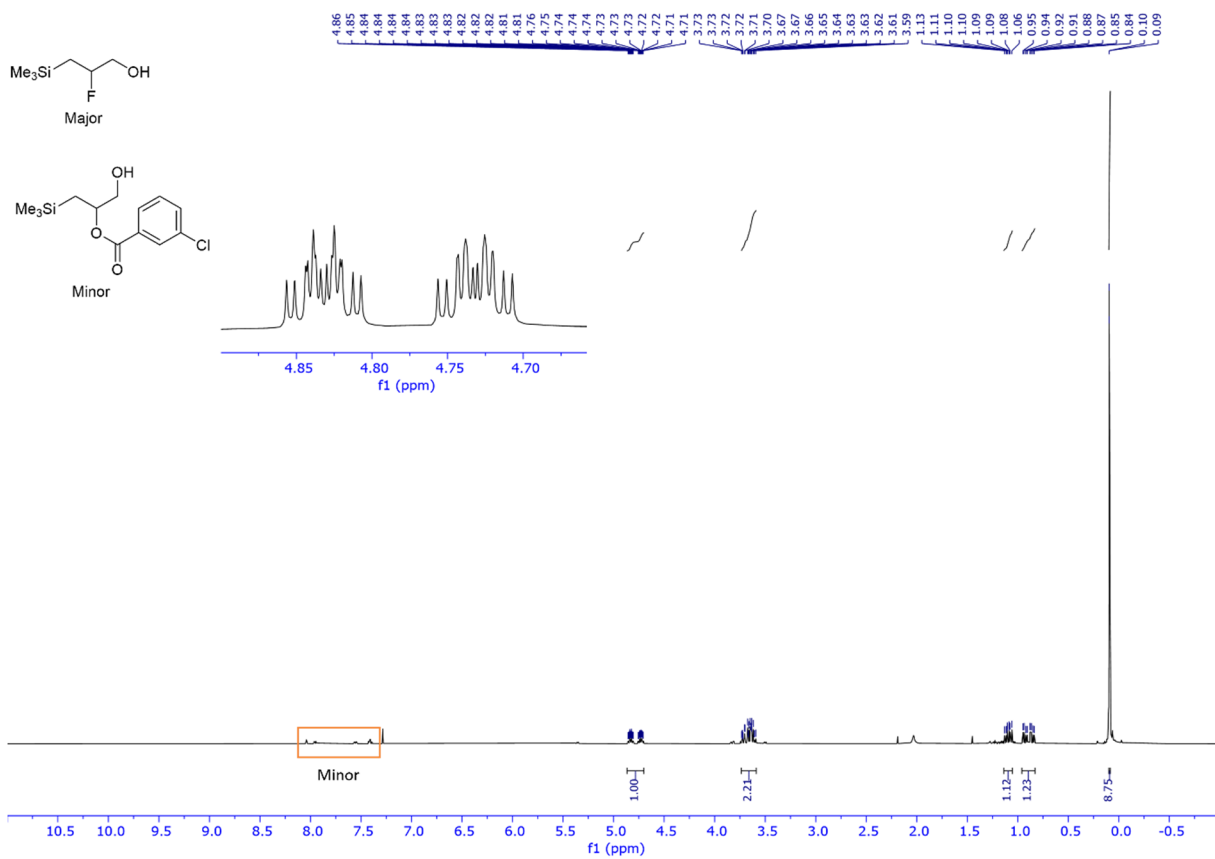
^{19}F NMR (470 MHz, CDCl_3) δ -173.78 (dddd, $J = 48.7, 33.3, 27.7, 20.6, 12.4$ Hz).

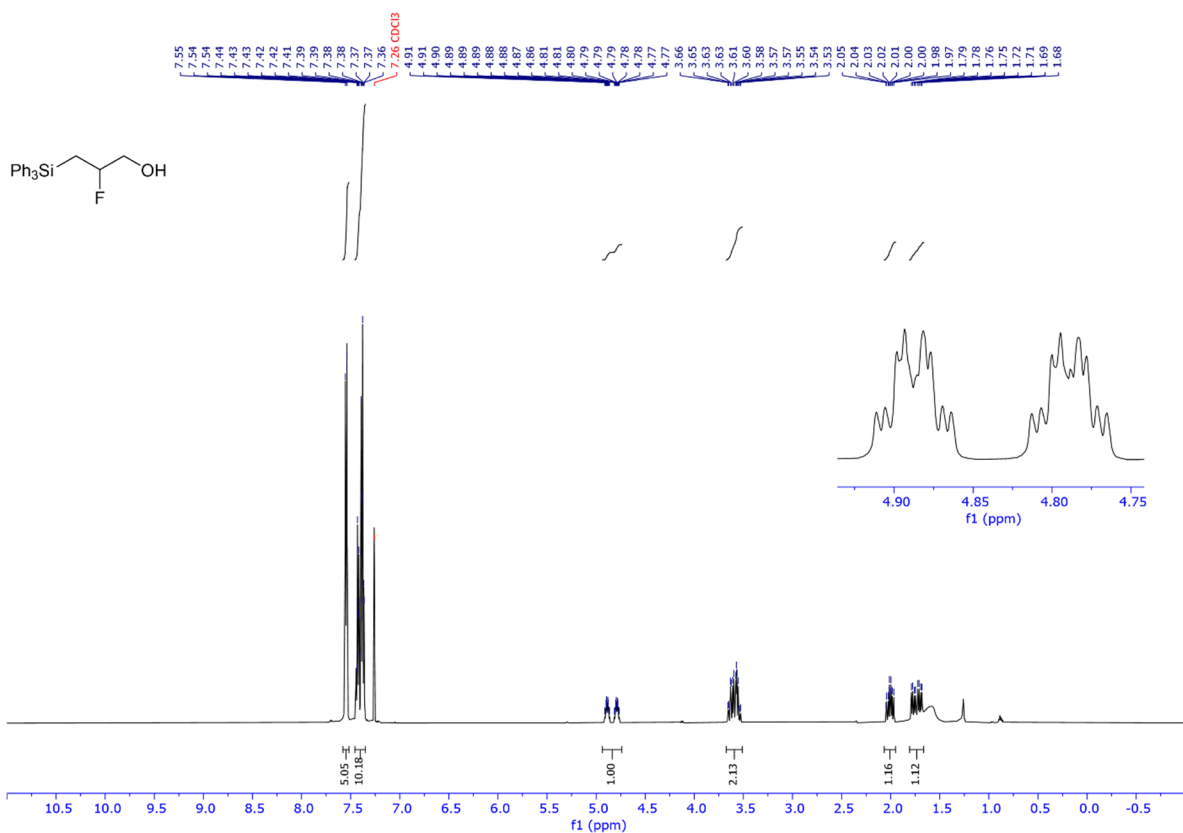
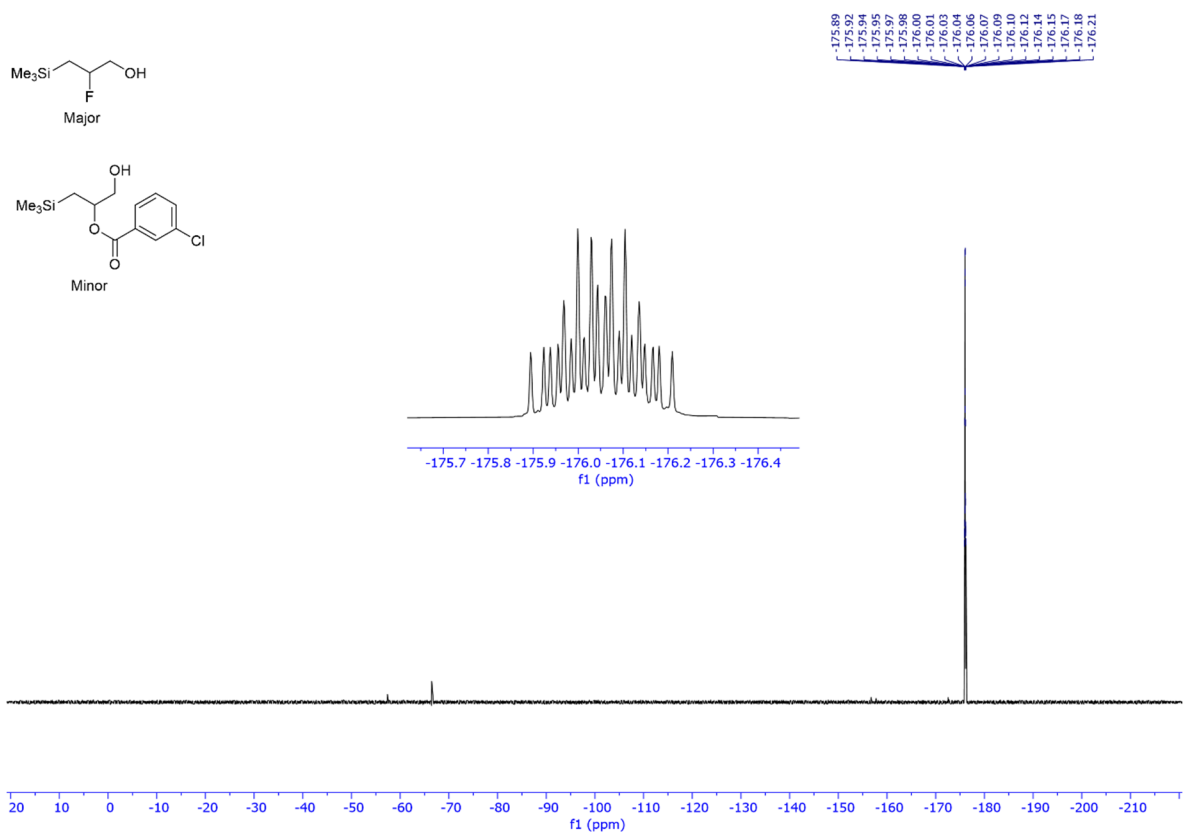
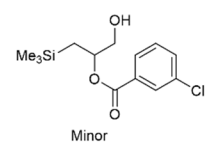
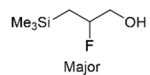
The pivaloyl ester (80.9 mg, 0.273 mmol, 1 equiv), KBr (48.7 mg, 0.41 mmol, 1.5 equiv), and NaOAc (67.7 mg, 0.82 mmol) were dissolved in acetic acid (1.3 mL) and the mixture was cooled to 0 °C. Additional NaOAc (178.6 mg, 2.17 mmol) was added followed by peracetic acid (0.96 mL, 4.7 mmol) and the mixture stirred for 15 minutes before adding additional peracetic acid (2.2 mL, 9.8 mmol). The flask was removed from the ice bath and allowed to warm while stirring for 1 hr. The reaction was carefully quenched by slowly adding aq. $\text{Na}_2\text{S}_2\text{O}_3$ (6 mL). Additional aq. NaHCO_3 (40 mL) was added and solid NaHCO_3 was added until effervescence ceased. Upon completion, the reaction mixture was extracted with CH_2Cl_2 (3 x 20 mL). The combined organic extracts dried with MgSO_4 and filtered before removing the solvent under reduced pressure. The crude product was purified by flash column chromatography on silica (4:1 hexanes:ethyl acetate, $R_f \sim 0.17$) to yield the compound as a colorless liquid (15.4 mg, 0.086 mmol, 32%).

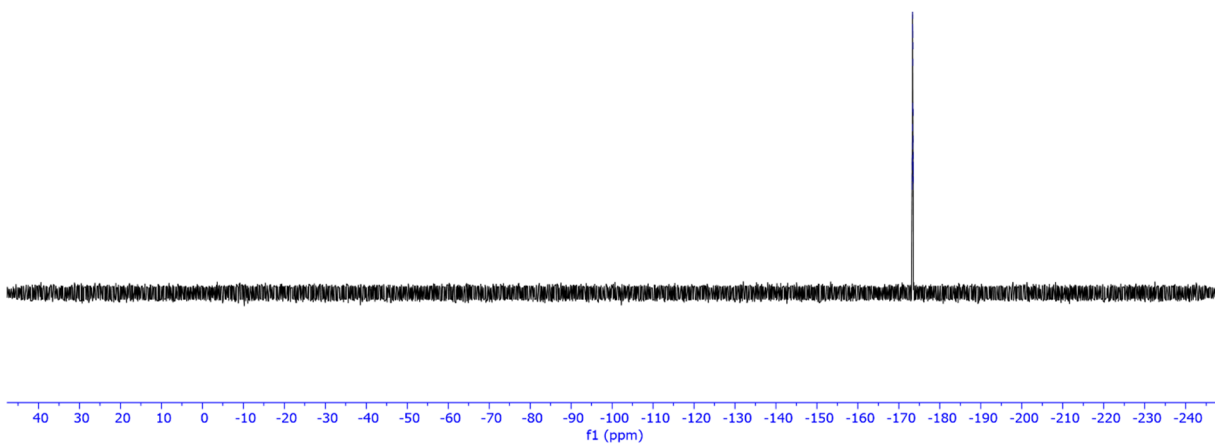
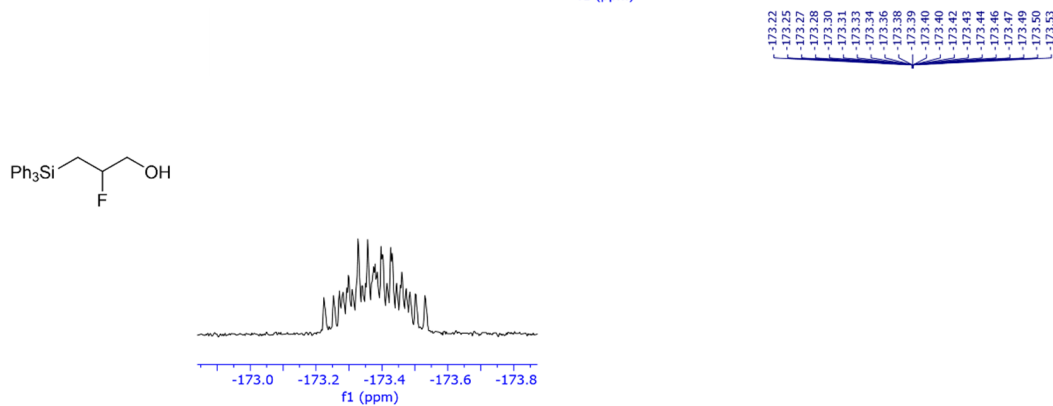
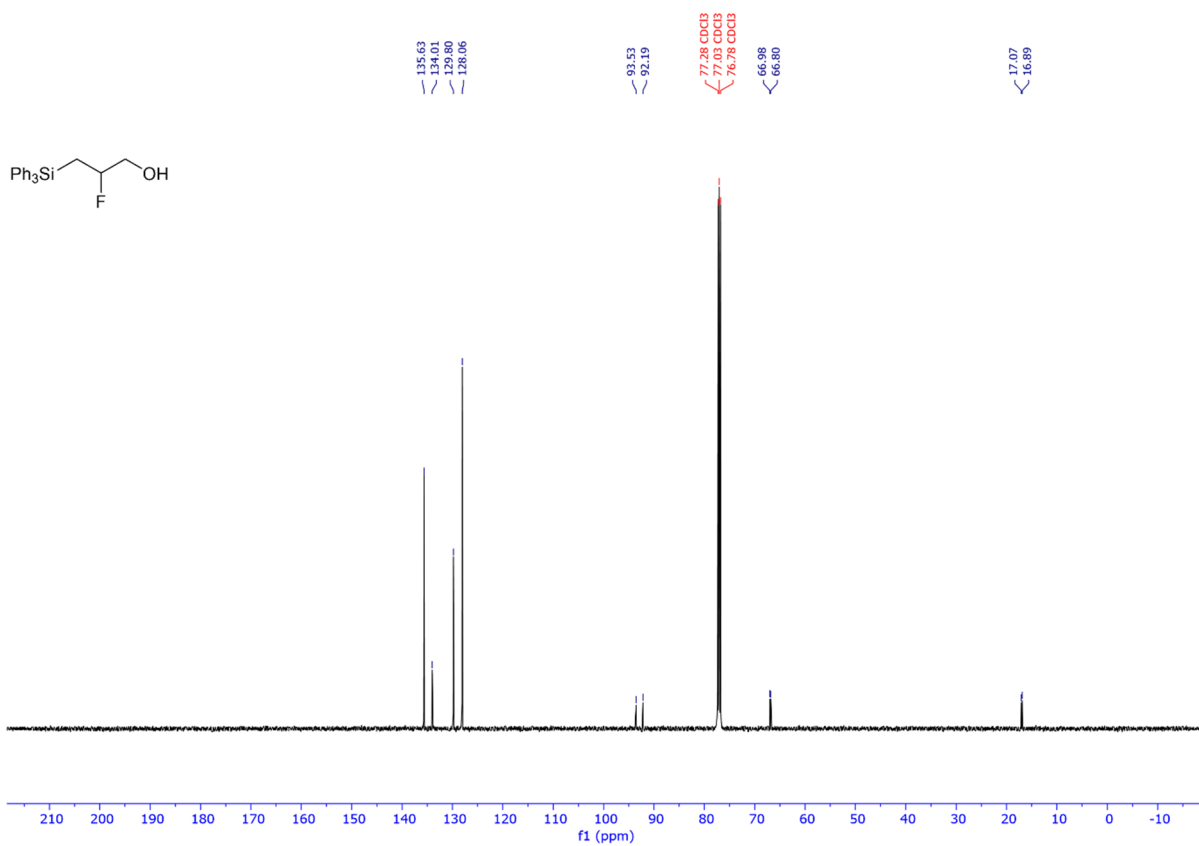
^1H NMR (500 MHz, CDCl_3) δ 4.74 (dq, $J = 47.9, 4.9, 4.1$ Hz, 1H), 4.37 – 4.25 (m, 2H), 3.79 (dd, $J = 21.4, 4.8$ Hz, 2H), 1.22 (s, 9H).

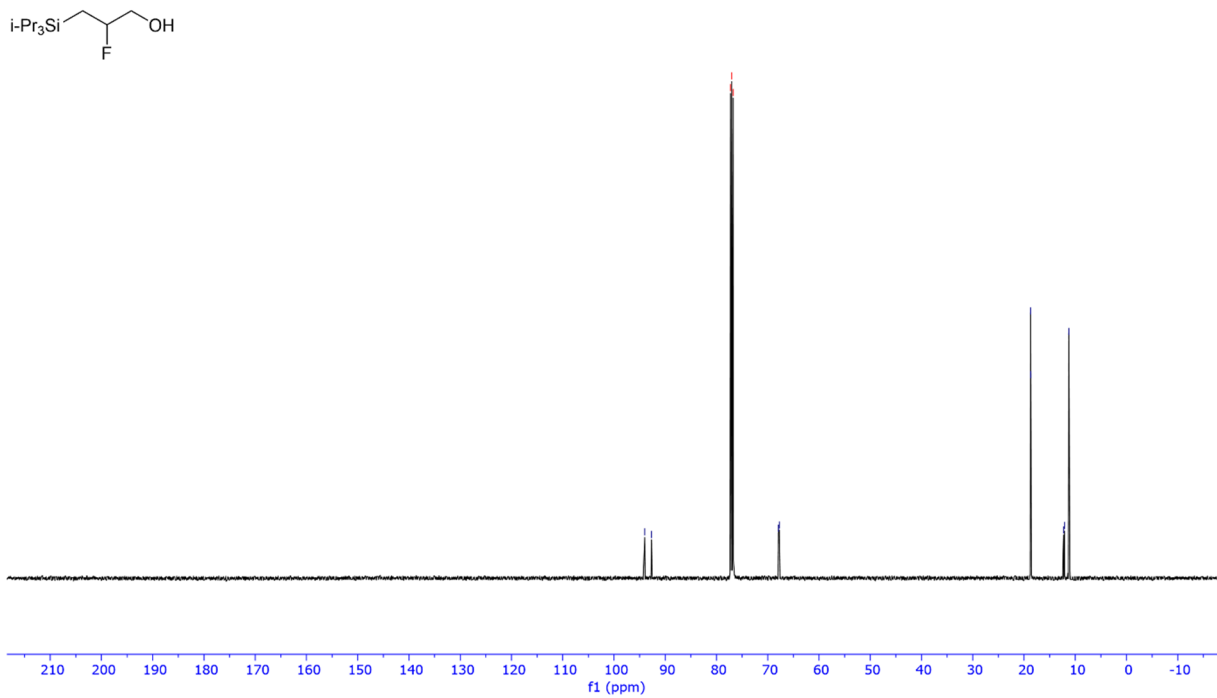
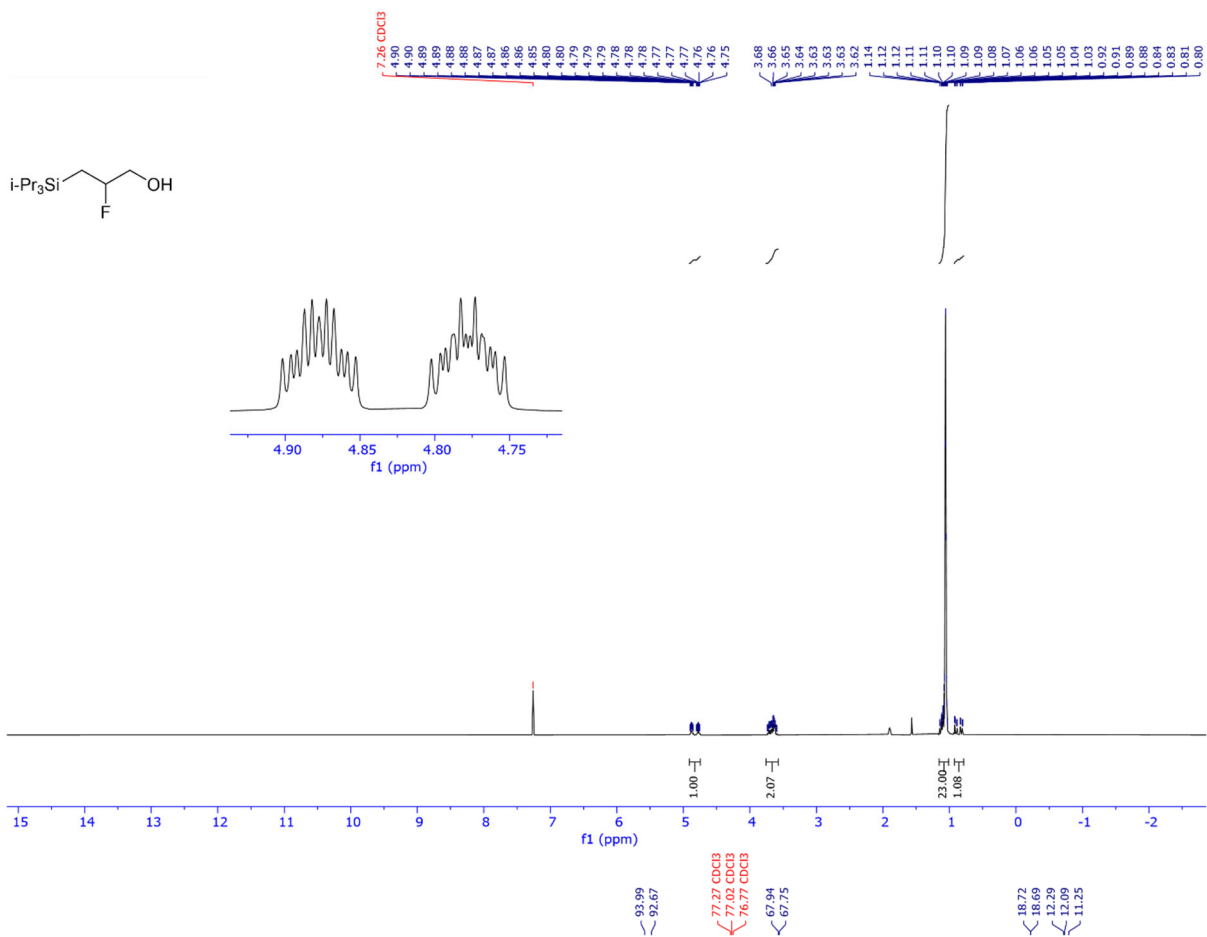
^{13}C NMR (126 MHz, CDCl_3) δ 178.52, 91.18 (d, $J = 173.5$ Hz), 62.54 (d, $J = 24.0$ Hz), 61.75 (d, $J = 23.2$ Hz), 38.89, 27.12.

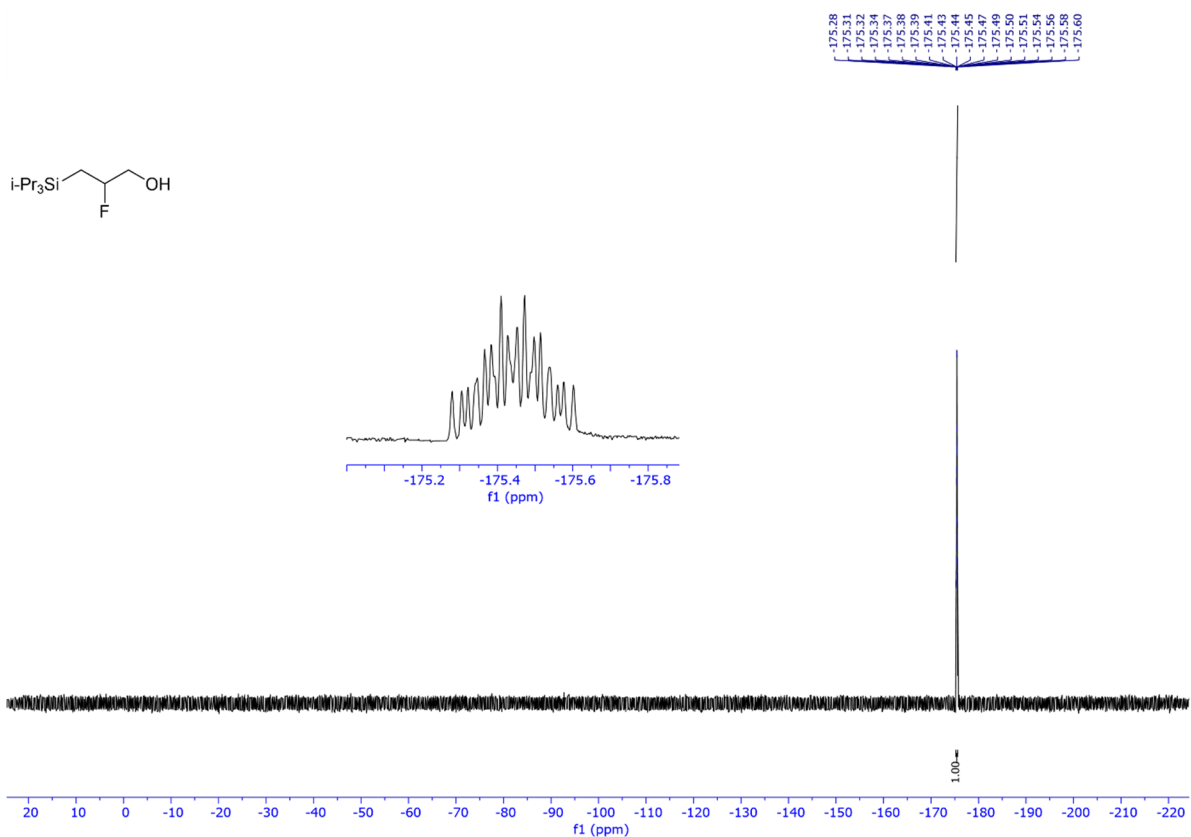
^{19}F NMR (470 MHz, CDCl_3) δ -197.46 (dq, $J = 48.3, 21.6$ Hz).

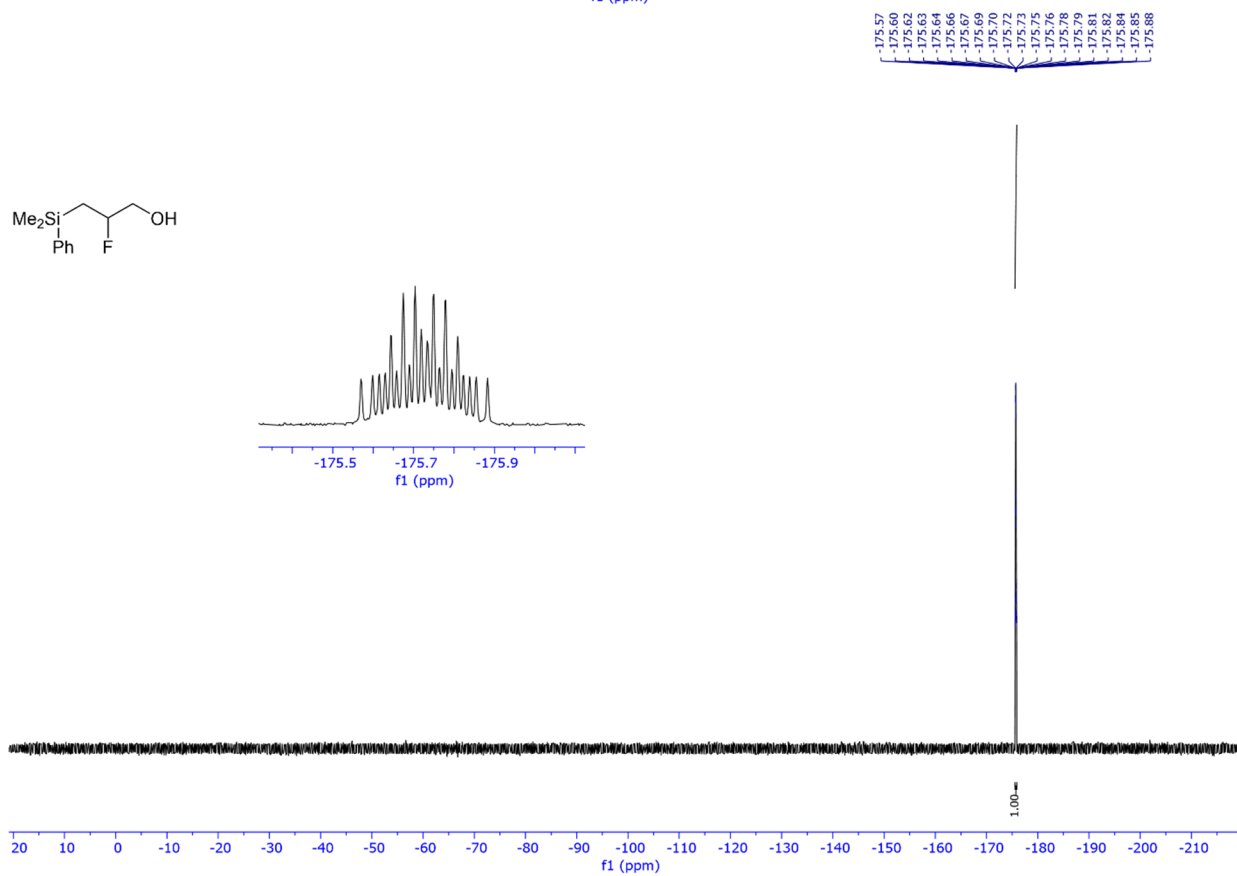
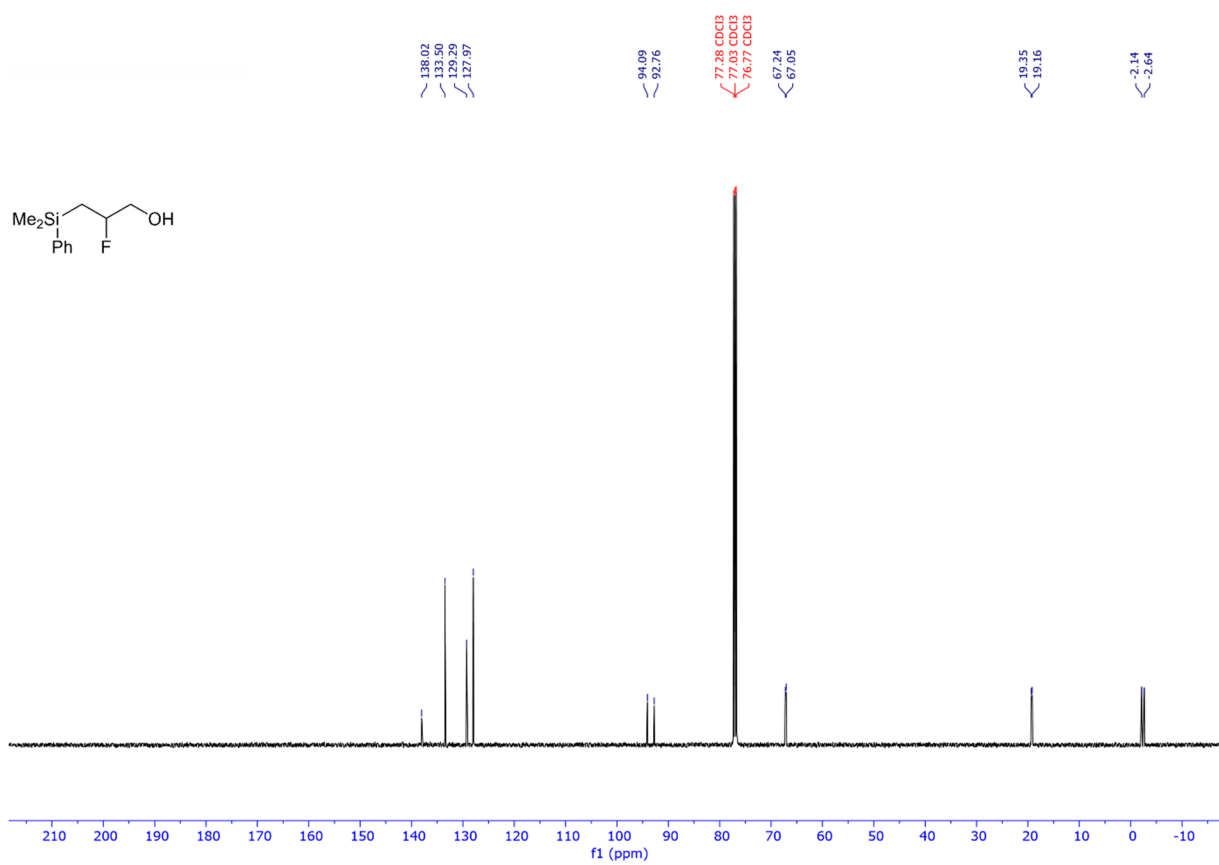


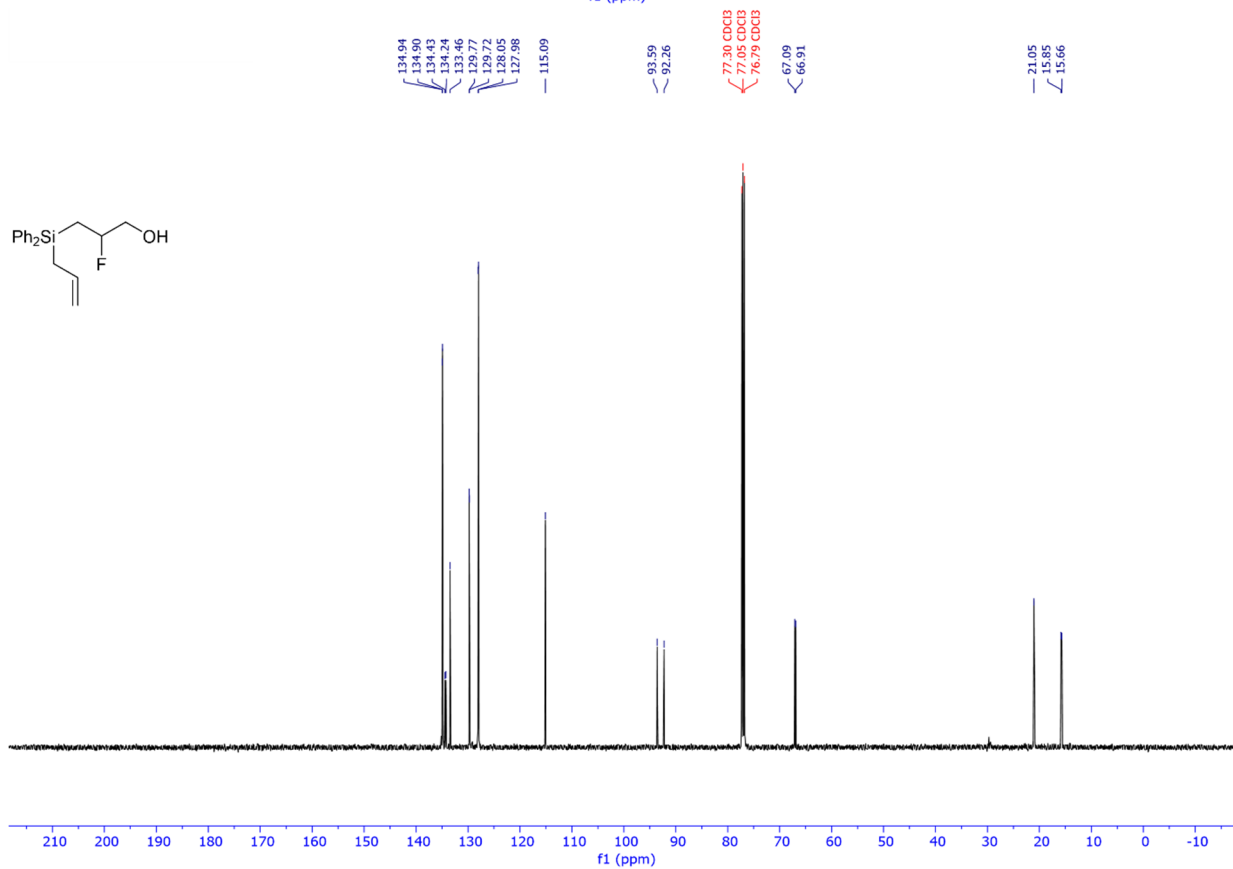
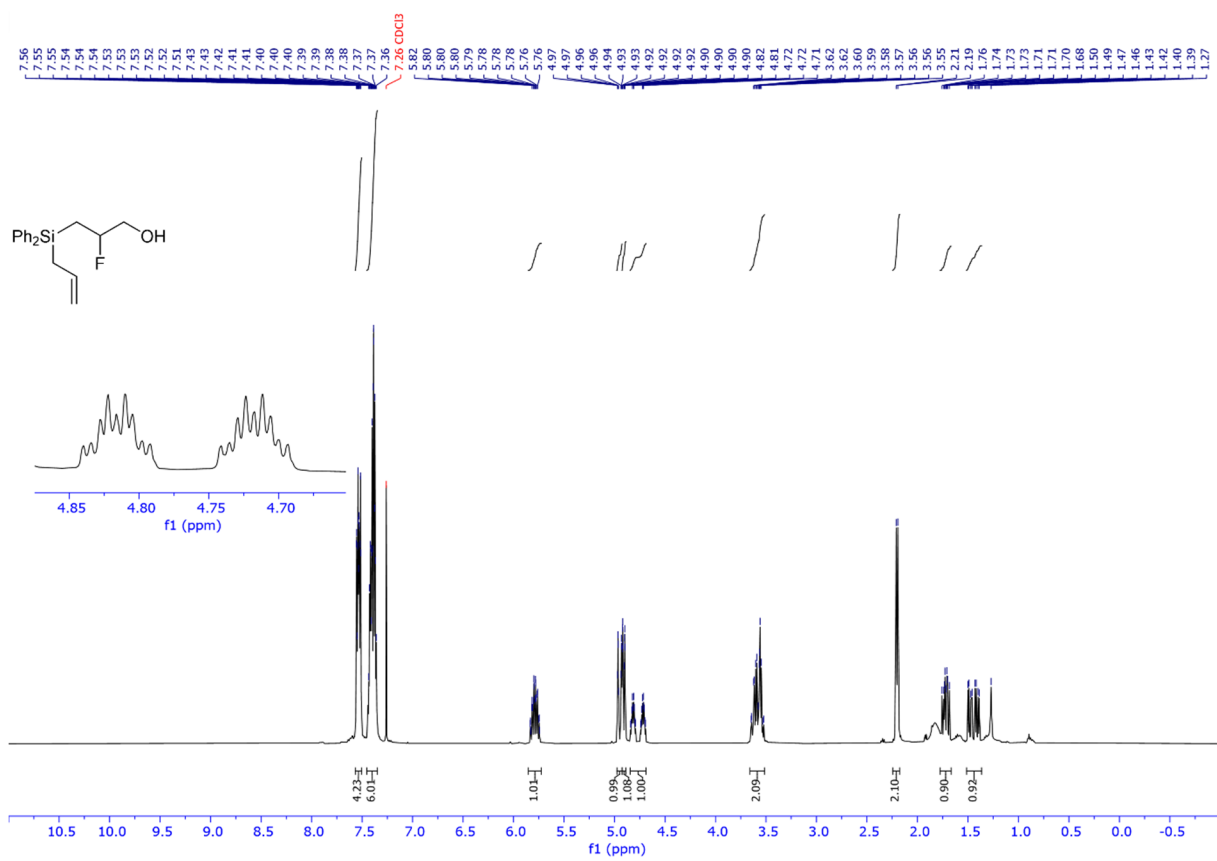


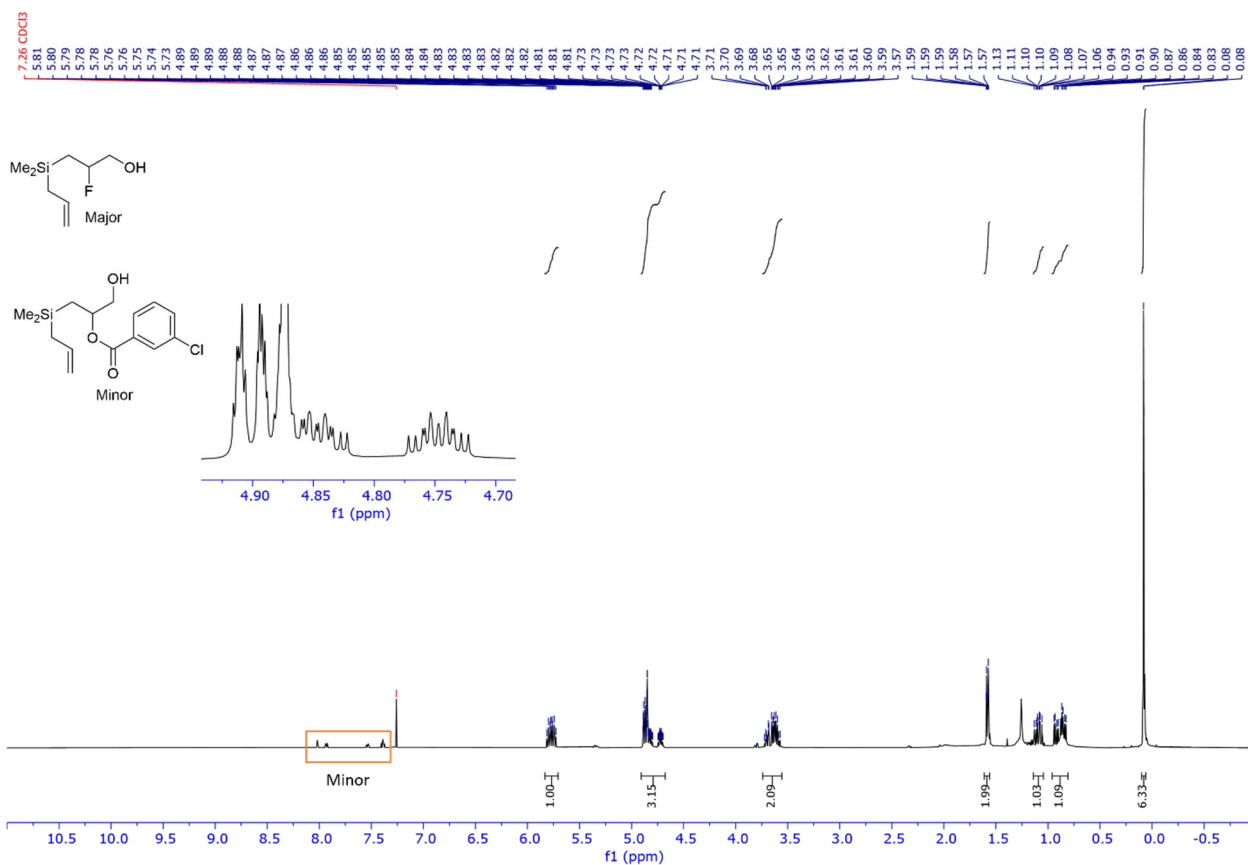
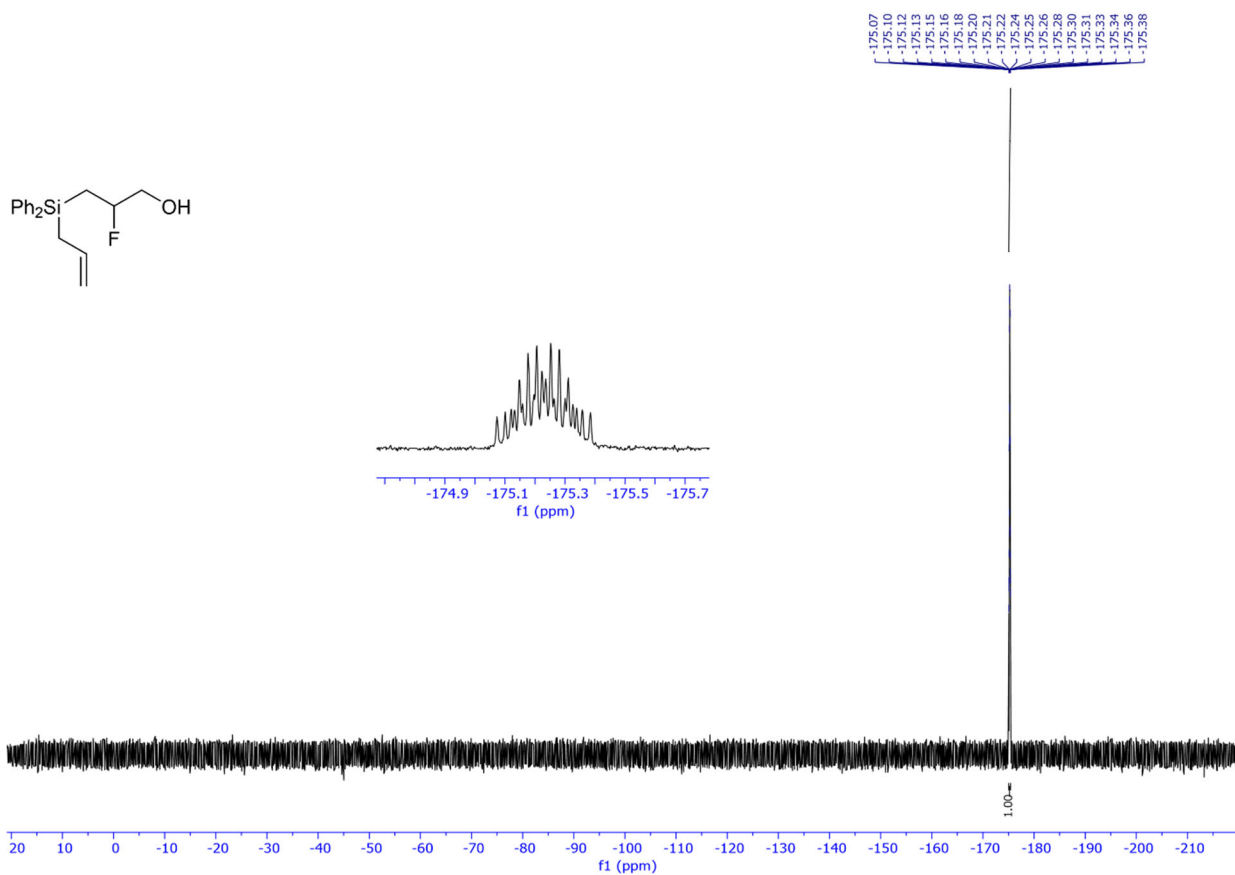
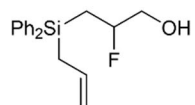


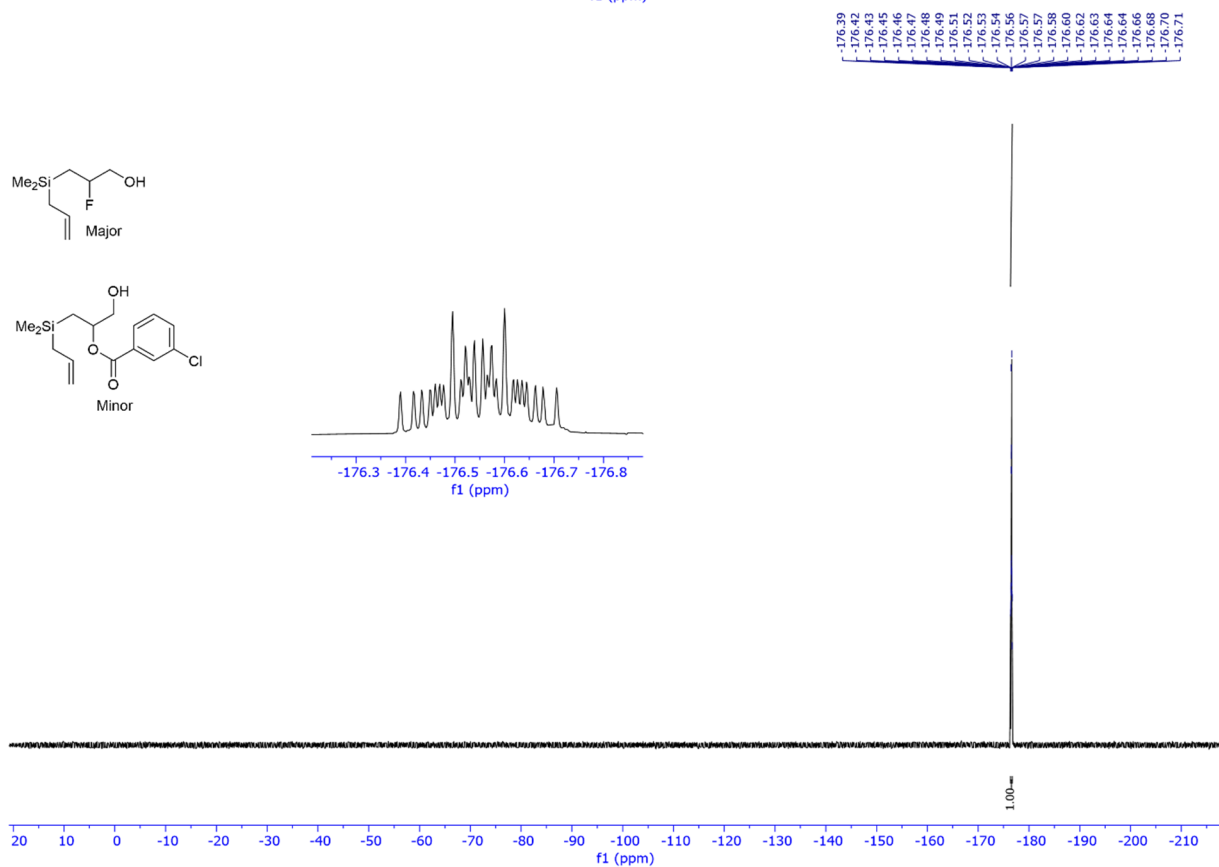
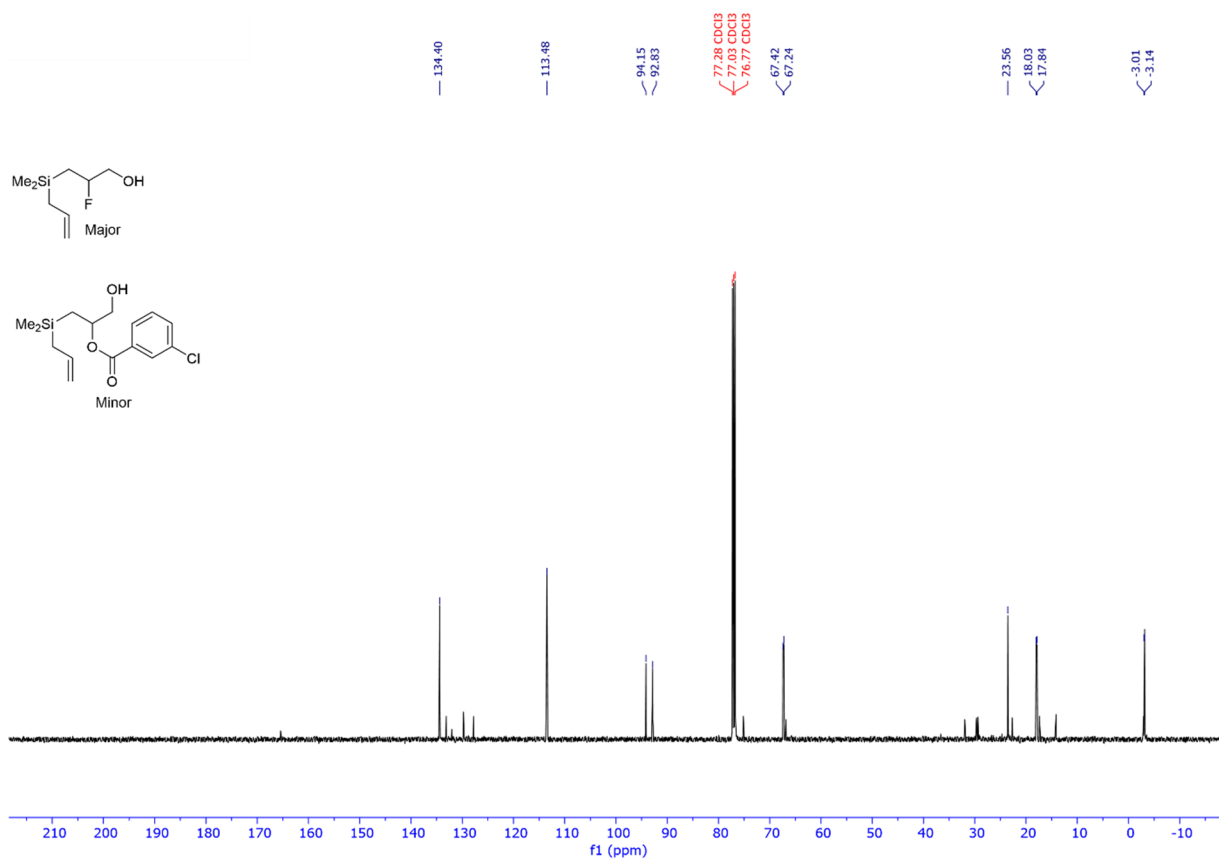


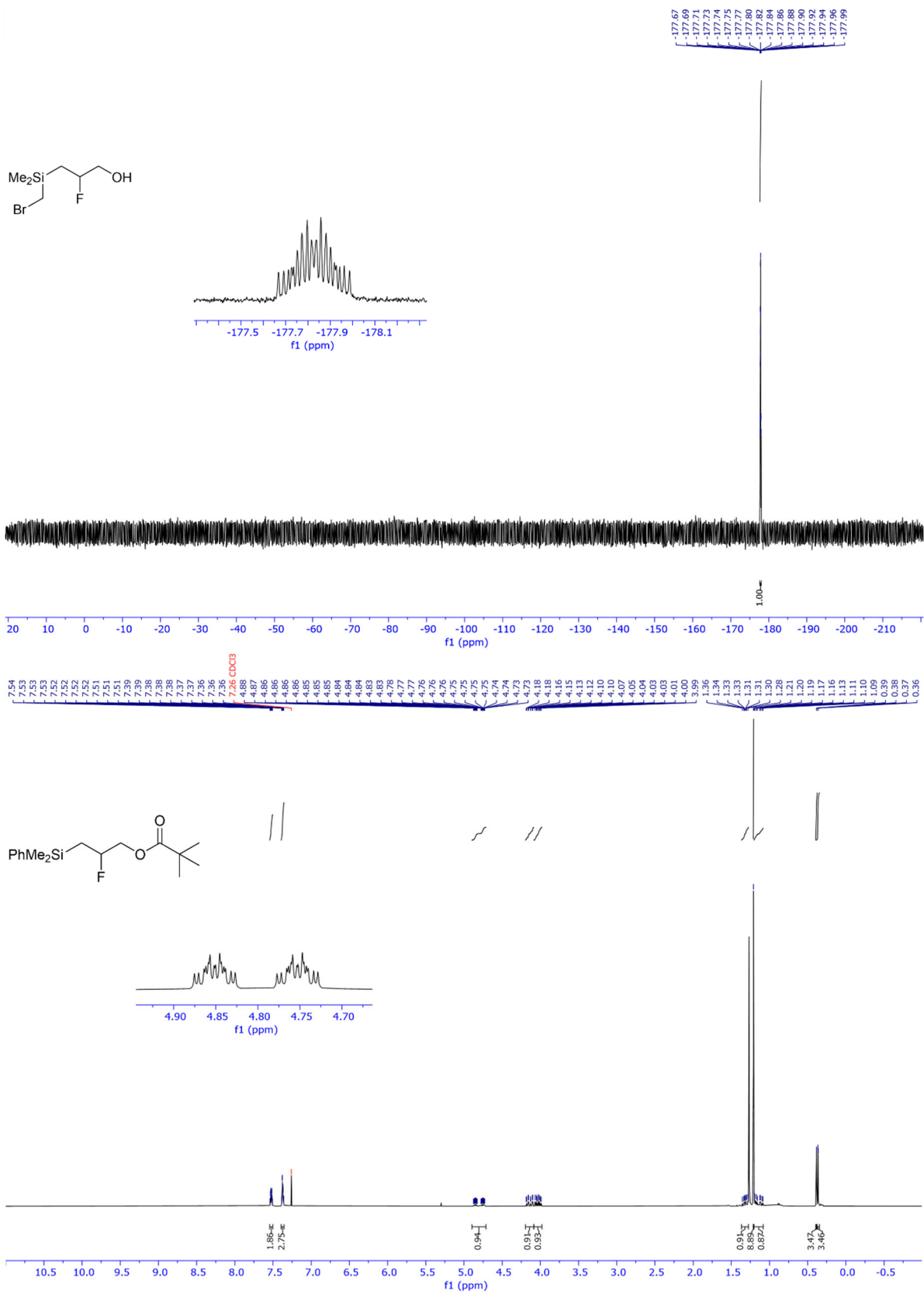


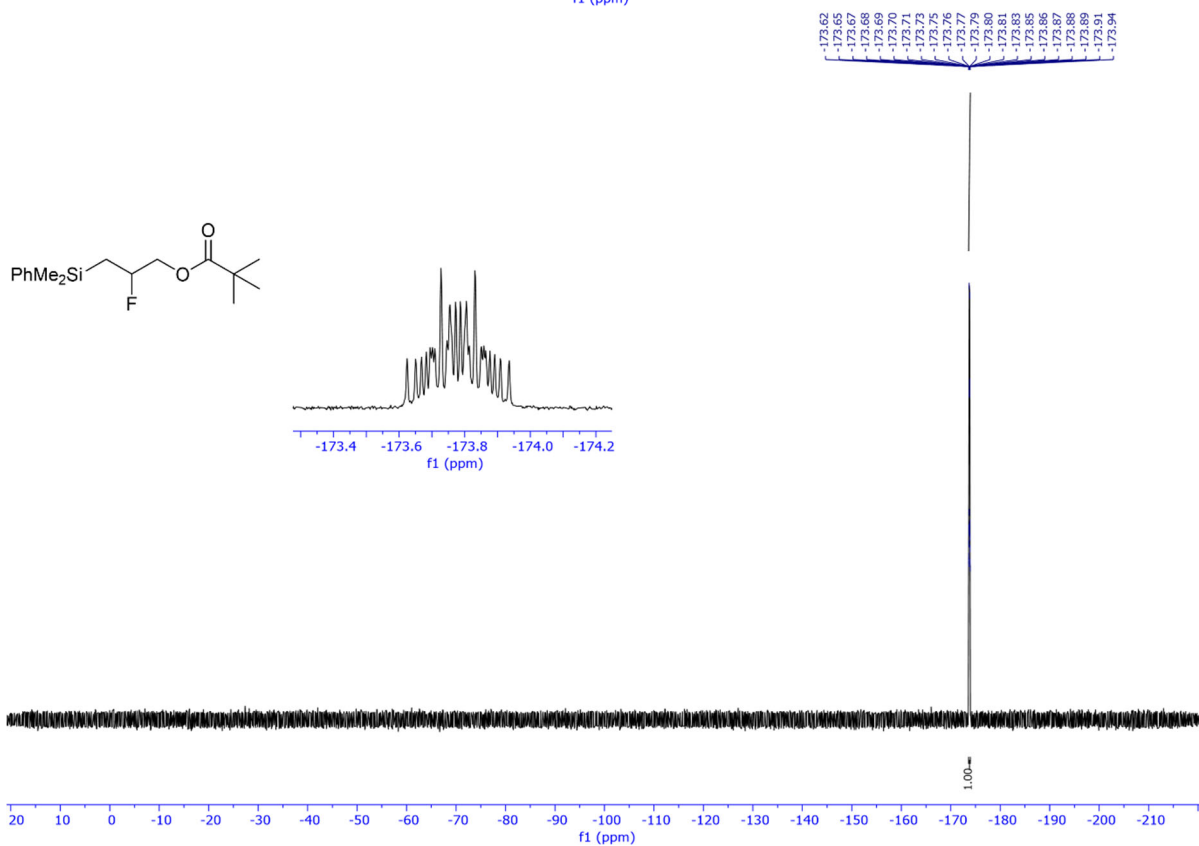
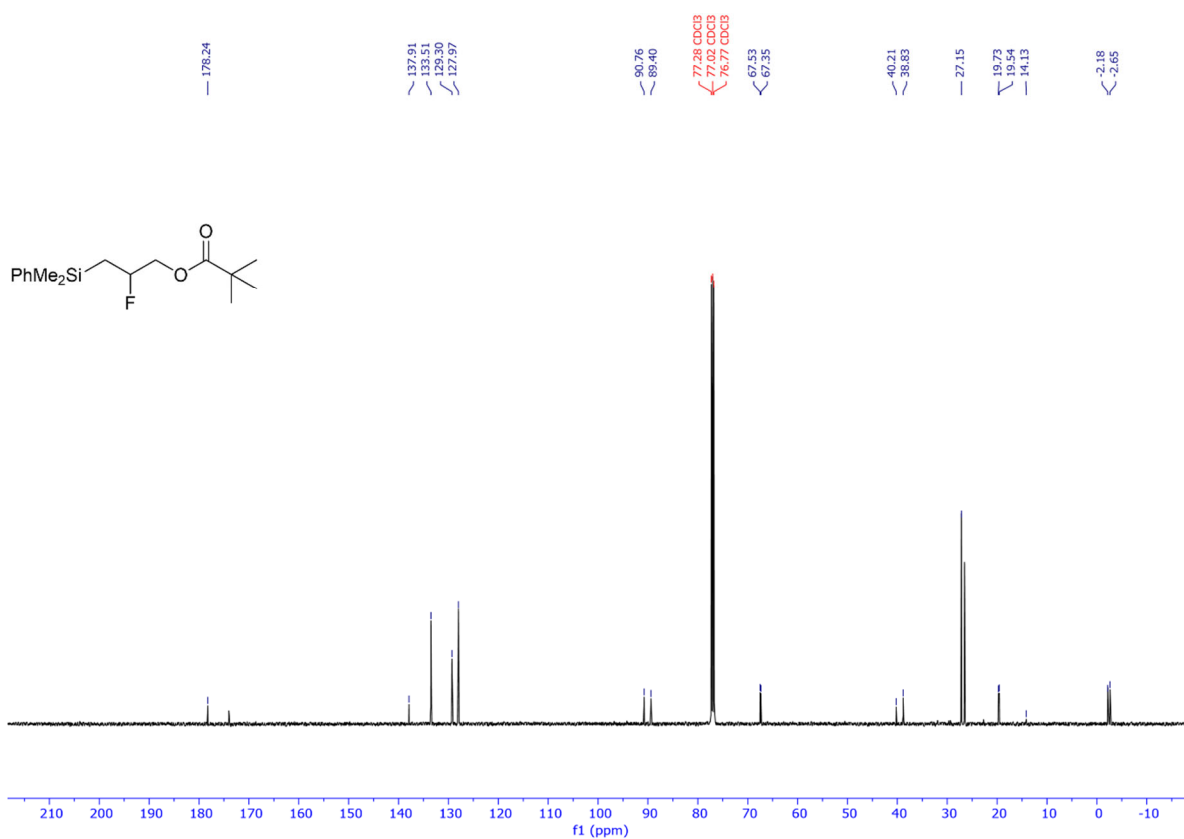


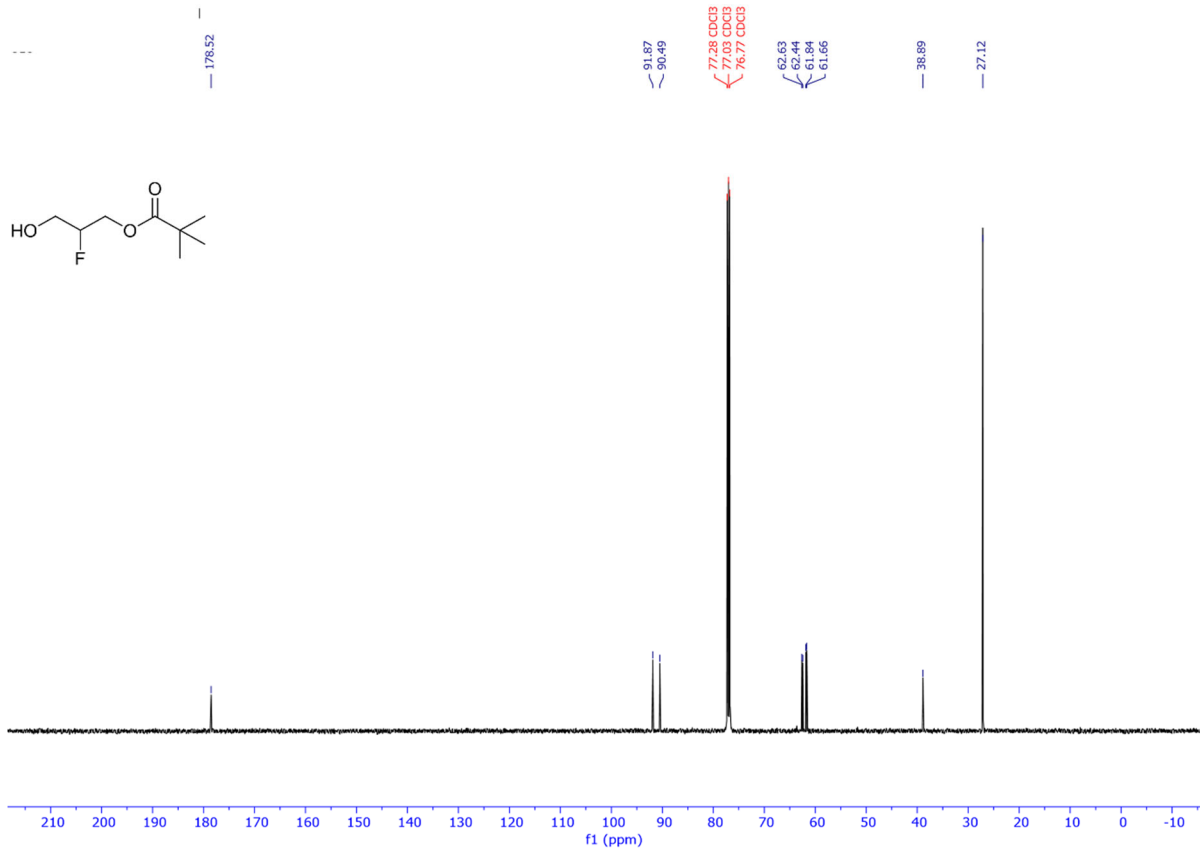
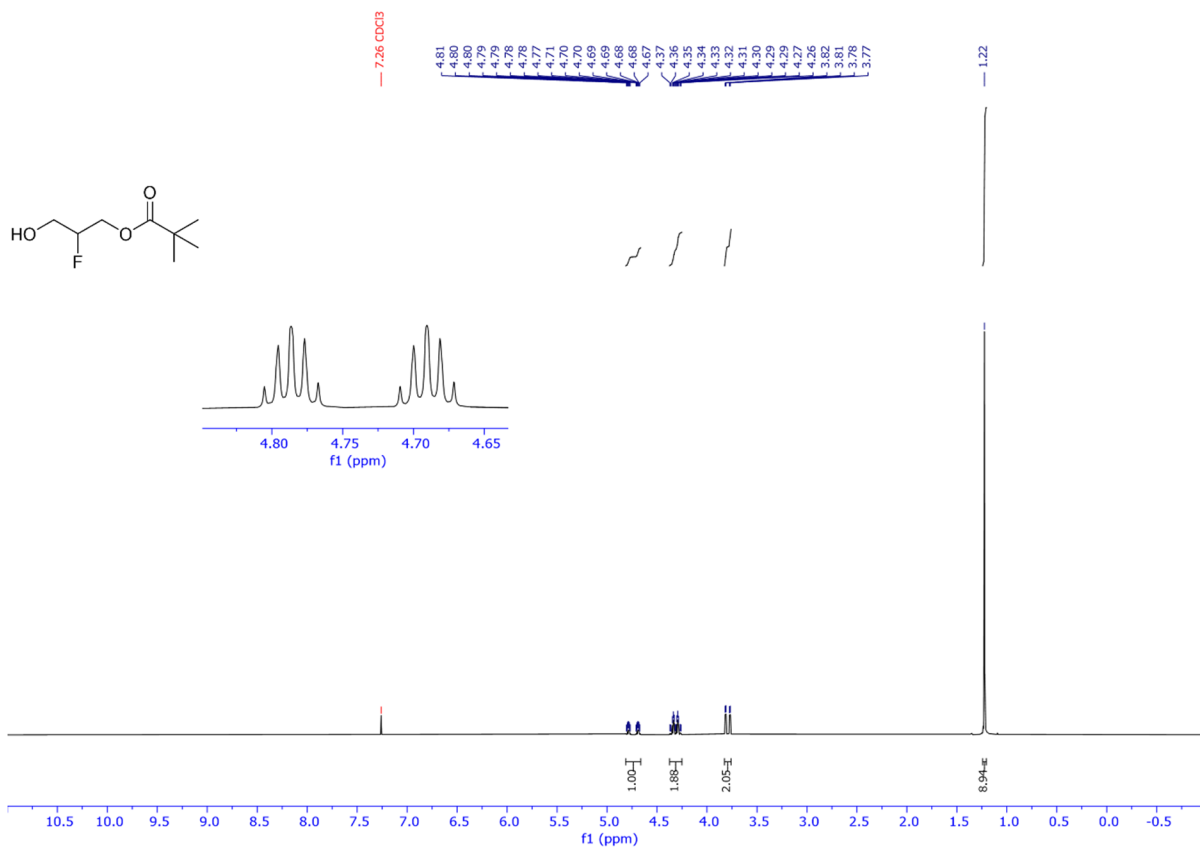












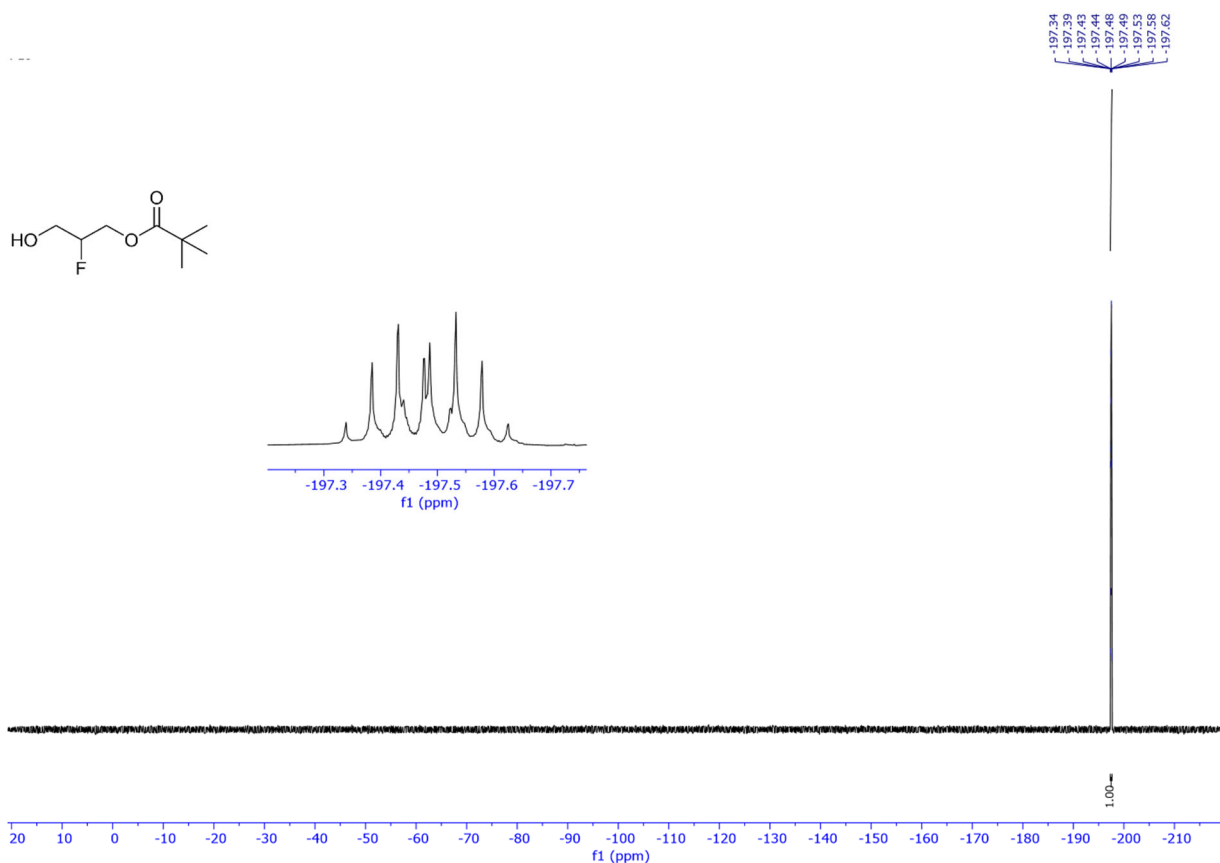


Figure S1. Mosher ester ^1H NMR analysis of enantioenriched 3-silylfluorohydrins. Integration values for the α -fluoro hydrogen in all cases was $\sim 1:5:1$, consistent with the *ee* measured for the triisopropylsilyl epoxide (22%)

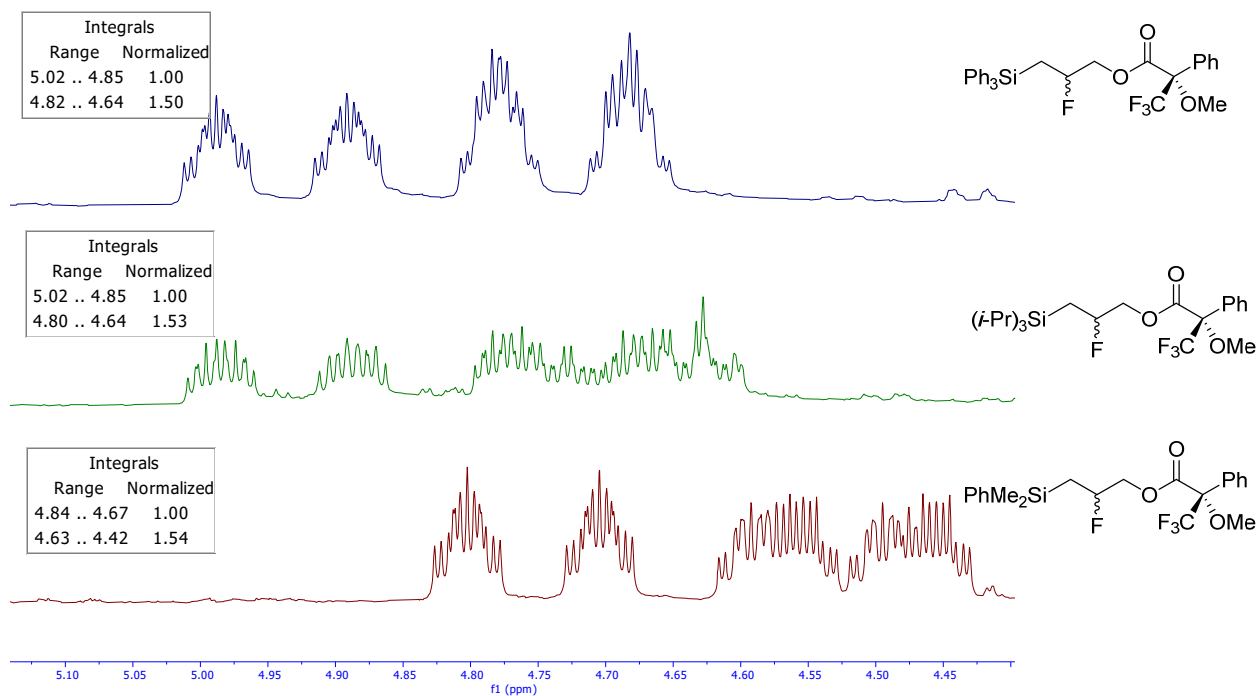


Figure S2. COSY NMR spectrum of compound **6**.

